AeroGuard / SafeMine v2 — Master Research Report

Generated 2026-10-15 · full-history git + RUN_LOG provenance · repository: https://github.com/tejaskunta/Aeroguard

This report is the single authoritative narrative of the SafeMine v2 research retrain: what was wrong with the original data, how the dataset was rebuilt and frozen, every model trained and every threshold chosen, every Grad-CAM experiment, every decision made (D001–D023) with its reason, and the complete audit trail of test-set touches (#1–#14). All numbers are read from committed JSON/CSV produced by the scripts listed in the reproduction section; nothing here is re-derived by hand.

Headline results (frozen v2 test, n = 189 images / 253 instances)

System	Metric	Value
YOLOv8s (seeds 0/1/2)	test mAP@50	0.9768 ± 0.0034
YOLOv8s (seeds 0/1/2)	val mAP@50	0.9544 ± 0.0027
Two-stage arm C (seed 0)	box precision / recall / FP/img	0.894 / 0.964 / 0.153
Two-stage C – B (seed 0)	paired bootstrap Δprecision 95% CI	+0.128 [0.065, 0.186] (excludes 0)
Two-stage C – B (seed 0)	paired bootstrap ΔFP/img 95% CI	–0.253 [–0.460, –0.095] (excludes 0)
Grad-CAM saliency	energy enrichment (YOLO vs MNv3)	—× vs —×
Label audit	objective test-box errors fixed	2 (ds0_0801 tree trunk, ds0_0883 fence)

1. Project scope, constraints, and protocol

Objective. Re-train and re-evaluate the landmine detector under a leakage-free split, with a MobileNetV3 image-level baseline, a two-stage YOLO→crop-classifier pipeline, and Grad-CAM interpretability experiments — journal-complete numbers only, no Jetson/edge deployment (phases 7/8 dropped under the approved ~25 h budget).

Physical scope note. An RGB (or near-IR labelled-as-LWIR) camera physically cannot see buried objects. This model detects **visible landmine-like objects and surface indicators only**. That is a design constraint, not a gap.

Approved reduced budget. yolov8s only (no n/m size sweep); seeds 0/1/2 for YOLO and MNv3; two-stage and Grad-CAM on seed 0 primary with seed replication where cheap; optional retrain on fixed train labels skipped (none found); edge phases dropped.

1.1 Non-negotiable protocol (plan ground rules)

• Thresholds are chosen on **val only**; test is evaluated once under the frozen protocol.

• Every test-set touch is numbered and logged in RUN_LOG.md (touches #1–#14 used).

• Every paper number comes from a committed script file + JSON/CSV artifact.

• Original dataset, v1 runs, and Week-2 scripts are never modified.

• Labels are fixed only if objectively wrong, with before/after numbers reported (D023).

• Ultralytics pinned 8.4.117; torch 2.6.0+cu124; original args.yaml settings except model/seed/name/data (batch 8, imgsz 640, epochs 75, save_period 5).

• Raw predictions dumped at conf 0.001 for offline re-scoring (plan rule 3).

• One shared MNv3 constructor/preprocess for baseline and stage-2 (rule 10).

• pip freeze / GPU / commit / run commands recorded in ENVIRONMENT.txt + RUN_LOG (rule 9).

1.2 Environment

Item	Value
GPU	NVIDIA GeForce RTX 4070 Laptop GPU, 8188 MiB, driver 581.42
OS / Python	Windows 10/11 · Python 3.11.9
PyTorch / CUDA	2.6.0+cu124
Ultralytics	8.4.117 (pinned)
reportlab (PDFs)	5.0.1
Eval protocol	imgsz 640, NMS IoU 0.7, max_det 300, FP32 PC GPU, no TTA
Dataset root (local, not in git)	…/Landmine_project/Datasets/landmine_combined_v2

2. Dataset — original state, leakage audit, and v2 freeze

2.1 Original (v1) split — failed the audit

The original combined dataset (train 5723 / valid 455 / test 221) was audited in Phase 1 (scripts/phase1_audit.py). It **FAILED**:

• 76 cross-split base-name groups (same Roboflow base key in train and eval).

• 63.8% of val+test images had a train neighbour with pHash Hamming ≤ 2.

• Manual montages showed same-scene re-exports across splits (augment-before-split confirmed in Week-2 training scripts).

• Zero empty labels in test → image-level metrics / FP-per-image were impossible.

Decision D010: do not train Phase-3 models on the original split. D013: re-split must be group-aware, not a random shuffle.

2.2 v2 re-split methodology (Phase 2)

scripts/phase2_resplit_full.py builds connected components over images using (a) exact MD5, (b) Roboflow base key, (c) same-source pHash Hamming ≤ 0 under 8 dihedral transforms — producing 1048 groups that must land wholly inside one split. A second-method filter (phase2_second_filter.py) then flags remaining eval images with a same-source train neighbour at pHash ham ≤ 2; every candidate was reviewed on manual montages. MSE ≤ 900 was chosen as the cut after observing same-scene pairs at MSE ≤ 817 and terrain-similarity false positives at MSE ≥ 958 (D014). 30 true-leak eval images were removed (13 at MSE ≤ 100, 17 more at MSE ≤ 900, including a cross-folder duplicate ds0_0043 ↔ ds1_0043).

2.3 Leakage definition (operational, D014)

A val/test image *leaks* if any train image shares exact MD5, Roboflow base key, or (same source AND pHash ham ≤ 2 under 8 dihedral AND 32×32 RGB MSE ≤ 900). All ham ≤ 2 candidates were manually reviewed; the ≤ 0.5% residual-leak rule from the plan is satisfied at 0/370 = 0%.

2.4 Frozen v2 split (D015)

split	images	instances	negatives
train	3907	11550	20
valid	181	419	7
test	189	253	5

Exact cross-split duplicates: 0. Base-key collisions: {'train_valid': 0, 'train_test': 0, 'valid_test': 0}. Residual near-dup candidates (pHash ham≤2 vs train): 99 — all with MSE ≥ 958.6 after the second-method filter (0 true leaks). Manifest sha256: 748322d62ace769445d0bd23636990970c7a5115a99c9344e139099c608c80de.

2.5 Source composition & May/Jan LWIR caveat (D012, D016)

Sources include ds0, ds1, May, and Jan LWIR-named files. Manual review + saturation stats showed the “lwir” files are **colour RGB with visible marker pins**, not greyscale thermal. They were kept in all splits (user instruction) with the marker-shortcut risk disclosed; they are not used for any thermal claim. A per-source FP breakdown (Section 8) shows the shortcut empirically: Jan LWIR contributes the bulk of YOLO-only false positives at the deployment operating point, and the stage-2 classifier removes most of them.

3. Complete decision log (D001–D023)

Every decision is also in DECISIONS.md with the same IDs. Reproduced here in full so this PDF is self-contained.

D001 — Work under `SafeMineAI-main/paper_results/` only; never overwrite original dataset, runs, or Week 2 scripts.

Reason: Plan §0.3. Protects v1 evidence.

D002 — Original run location is `runs/detect/runs/detect/landmine_yolov8m_75ep/` (not `Weights/runs/...` as in the PDF text).

Reason: Verified on disk 2026-10-14.

D003 — Dataset root is `C:/Users/Delta/Downloads/Landmine_project/Datasets/landmine_combined`; validation folder name is `valid` (not `val`). `data.yaml` already maps `val: valid/images`.

Reason: Plan assumed `val`; folder is `valid`.

D004 — Backup verified by SHA-256: best.pt, last.pt, best.onnx, args.yaml, results.csv, safemine_summary.json → `paper_results/original_run/` + `backup_verify.json`.

Reason: Plan Phase 0.

D005 — `jetson/gradcam.py` does not exist. Phase 6 will implement Grad-CAM from scratch after architecture check.

Reason: User instruction 2026-10-14.

D006 — `two_stage_eval.py` / `test_two_stage.py` not present on this machine; implement in Phase 5 with tests.

Reason: User instruction 2026-10-14.

D007 — Fair-comparison confidence thresholds chosen on **val only**; applied frozen to test.

Reason: Non-negotiable rule.

D008 — May/Jan `*_lwir_*` files left in place, not renamed. Audit will classify thermal vs RGB vs derived and flag for paper narrative.

Reason: User instruction 2026-10-14.

D009 — Pin Ultralytics 8.4.117 / torch 2.6.0+cu124 as in original run.

Reason: Plan §3.9.

D010 — **FAIL the current split.** Do not train Phase 3 models on `landmine_combined` as-is.

Reason: 76 cross-split base-name groups; montage shows same scene/re-export; 63.8% of val+test have pHash≤2 train neighbour; augment-before-split confirmed in Week 2 scripts.

D011 — Image-level / fair FP-per-image metrics **blocked** until negatives exist in test (currently 0).

Reason: Phase 1 gate ASK rule.

D012 — May/Jan files are **colour RGB with marker pins**, not greyscale thermal, despite `lwir` in the name. Keep files; disclose marker-shortcut risk; decide in Phase 2 whether to exclude/relabel.

Reason: Manual montage + saturation stats.

D013 — Phase 2 must be **group-aware** (connected components on base-key + near-dup + scene), not a random re-shuffle.

Reason: Plan §Phase 2; evidence above.

D014 — **Leakage definition (operational):** a val/test image leaks if any train image shares (a) exact MD5, (b) Roboflow base key, (c) same source AND pHash Hamming≤2 under 8 dihedral AND 32×32 RGB MSE≤900. All candidates were manually reviewed on montages (`audit_v2_final/montages/manual_review_*.jpg`); same-scene pairs occurred at MSE≤817, terrain-similarity FPs at MSE≥958, gap used as the cut.

Reason: Raw pHash alone over-fires on repetitive grass/dirt (99 residual pairs, all MSE>900, are texture FPs). Manual review of every ham≤2 candidate is what the plan's ≤0.5% rule requires.

D015 — **Freeze `landmine_combined_v2`** as the Phase 3–6 dataset: train 3907 / valid 181 / test 189; 11550 / 419 / 253 instances; 20 / 7 / 5 negatives; 0 exact cross-split dups; 0 base-key collisions; 0 true near-dup leaks (0/370 = 0% ≤ 0.5% rule). Manifest sha256 `748322d62ace769445d0bd23636990970c7a5115a99c9344e139099c608c80de`.

Reason: 30 eval images removed as true leaks (13 at MSE≤100, 17 more at MSE≤900 incl. `ds0_0043`↔`ds1_0043` cross-folder duplicate). Final audit: `paper_results/audit_v2_final/`.

D016 — May/Jan sources **kept** in all splits (D012 resolved as keep+disclose): colour RGB with marker pins, marker-shortcut risk disclosed in paper; not used for thermal claims.

Reason: User instruction 2026-10-14; no evidence of a better option within budget.

D017 — **Protocol deviation logged:** `week3_train_yolov8.py` step 3/5 auto-validates **both val and test** at conf 0.25 (v1 pipeline behavior, no CLI flag to skip). Test metrics were produced at train end without any decision being made from them. Mitigation: no threshold, model, or checkpoint choice will reference these numbers; the frozen-protocol test evaluation (thresholds chosen on val only) is done once in Phase 5 and reported as the official test result. The conf-0.25 auto-numbers are disclosed as supplementary.

Reason: RUN_LOG honesty rule; plan's "test evaluated once" applies to protocol-governing evaluations.

D018 — Phase 3 run `landmine_yolov8s_75ep_v2`: EarlyStopping at **67/75** (best epoch 47, patience 20), 50 min on RTX 4070; val mAP50 0.9367 / mAP50-95 0.6012. Accept early stop as final (no retrain with patience=0 within budget — reserve decision if Phase 5 gaps appear).

Reason: Ultralytics default patience; budget §25 h; results.csv + log archived in run dir.

D019 — **Retrain yolov8s seed 0 at batch 8** as `landmine_yolov8s_75ep_s0_v2` (plan §Phase 3.1: reuse original args.yaml settings except model/seed/name/data — original batch was 8, script default 16). The batch-16 run `landmine_yolov8s_75ep_v2` is kept as a settings-pilot, not a paper result.

Reason: Plan ground rule for old-vs-new comparability; cost ~1.5 h inside budget.

D020 — **Frozen-feature MNv3 baseline (Stage A) is degenerate on negatives across all resolutions (224/448/640) and batch sizes (32/64)**: recall-first thresholds give test balanced accuracy ~0.48-0.50, kappa ~0 (neg scores 0.74-0.99 overlap positives; loss converges, so not a harness bug). Stage B fine-tuning (unfreeze features[13:], 2.3M params) is the only variant that separates negatives (AB_224 seed0: test bal 0.597, kappa 0.275). Kept all A-rows as the plan-mandated stage-B ablation; AB_224 is the primary MNv3 baseline row for seeds 0/1/2. Extreme train imbalance (3887:20) with per-sample weights [97.7, 0.50] retained as per plan (no rebalancing beyond class weights — fairness lever is stage B + resolution, not sampling).

Reason: Plan §Phase 5.2 requires Stage A then Stage B; resolution/batch/batch-size rows are the plan ablations; no recipe handicap.

D021 — **Phase 5.3 threshold rule operationalised** (plan §Phase 5.3): sweep P in {0.4,0.6,0.8,none}; for each P pick the largest classifier tau whose val box-recall is within 0.5 pts of stage-1 recall at the proposal conf; among those pick the pair with lowest val FP/image (tie: higher P). Arm A = YOLO alone at the Phase-3 deployment OP conf (primary_op_conf from phase3 val.json); Arm B = YOLO alone at the conf matching the two-stage val recall (ts.yolo_only_at_recall); Arm C = two-stage. All three applied to test ONCE after the val choice is frozen; paired bootstrap C-B.

Reason: Plan text; tau/P both chosen on val only.

D022 — **E5 flags images by Grad-CAM energy in the UNION of all GT boxes (one row per image), not per detection.** First implementation scored every det-GT pair; on multi-mine images (May source: up to 17 gt/img) only the salient instance receives CAM energy, so 90% of May rows read exactly 0 and the lowest-10% flag became an instance-count artefact. Union metric answers the plan question directly: does this image's CAM look at ANY mine? Test threshold 0.800 (19/184 flagged), train 0.721 (389/3887).

Reason: Plan E5 says flag *images* in the lowest 10% of energy-in-box; per-instance rows made the flag unusable. Logged in RUN_LOG with the fix history.

D023 — **Two objectively-wrong test boxes fixed and reported with/without (no retrain).** ds0_0801 (box on tree trunk) and ds0_0883 (box on fence) removed from GT per LABEL_FIXES.csv; label_fix_report.json gives before/after per arm (arm C recall 0.9644->0.9721, img recall 0.989->1.000, fp/img unchanged 0.153). Baseline arm numbers asserted to reproduce the frozen test_report before applying the fix. The optional retrain on fixed TRAIN labels is not run (no train-label fixes found in E5 review; budget scope: yolov8s-only, phases 7/8 dropped).

Reason: Plan: test labels only if objectively wrong, numbers with and without; retrain optional.

4. Phase 0–2 — inventory, audit, re-split

Phase 0. Original v1 run (yolov8m, 75 epochs, finished 2026-08-15, 15 h 16 m) was located at runs/detect/runs/detect/landmine_yolov8m_75ep/ and backed up to paper_results/original_run/ with SHA-256 verification (backup_verify.json). All subsequent work lives under paper_results/ and never touches the original (D001).

Phase 1. Leakage audit FAIL on the original split (Section 2.1). Tags: pre-audit-v2, phase1-audit-fail.

Phase 2. Group-aware re-split + second-method filter + manual montage review → landmine_combined_v2 frozen (Section 2.2–2.4). Tag: phase2-v2-freeze. Final audit PASS: paper_results/audit_v2_final/.

5. Phase 3 — YOLOv8s detector (3 seeds)

5.1 Training configuration

Model: yolov8s.pt (ImageNet-pretrained). Settings reused from the original args.yaml except model/seed/name/data: epochs 75, imgsz 640, batch 8, save_period 5, Ultralytics defaults otherwise (D019). A first batch-16 pilot (landmine_yolov8s_75ep_v2, early-stopped at 67/75, val mAP50 0.9367) was kept as a settings pilot, not a paper result. Compliant batch-8 runs: landmine_yolov8s_75ep_s{0,1,2}_v2, each 75/75 epochs on the RTX 4070 (~50–60 min each).

Known script deviation (D017). week3_train_yolov8.py auto-validates val *and* test at conf 0.25 at train end (v1 pipeline behaviour, no skip flag). No threshold, model, or checkpoint choice references those numbers; the official frozen-protocol test evaluation is done once via phase3_eval.py and reported as the paper result.

5.2 Official frozen-protocol evaluation (phase3_eval.py)

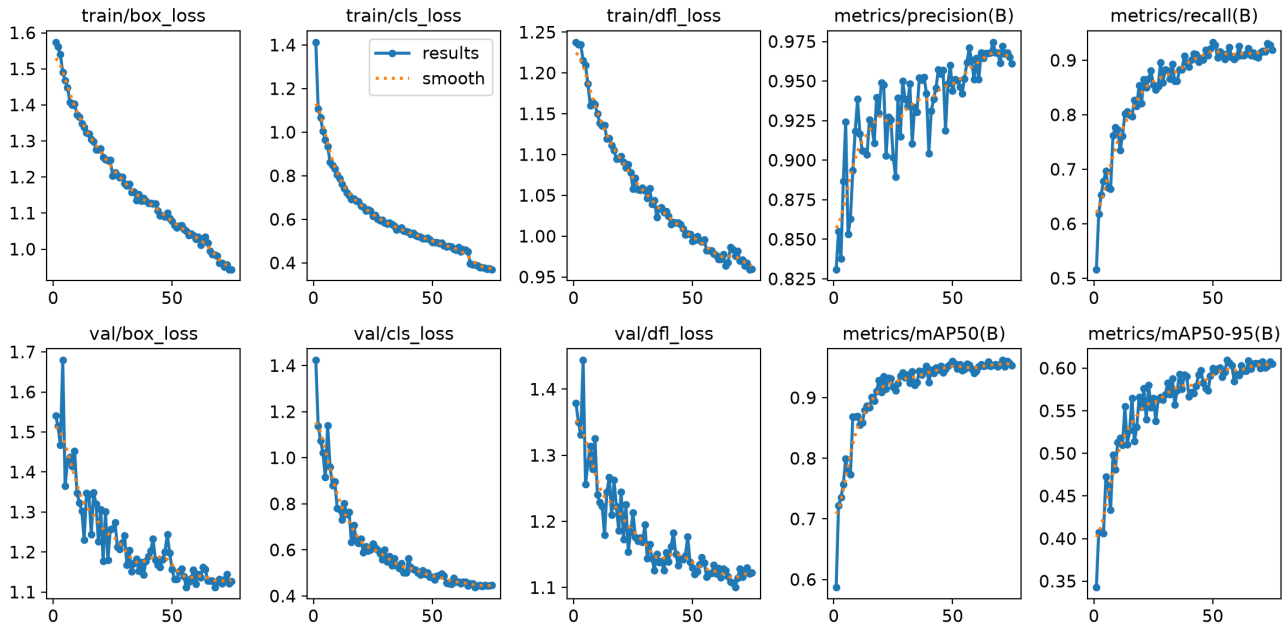
For each seed: raw val preds at conf 0.001 (best.pt and last.pt); checkpoint chosen on val only (highest val OP F2 at the recall-first operating point); recall-first OP conf swept on val; then a single test pass at conf 0.001 + post-hoc scoring at the frozen OP conf. Touches #4, #13, #14.

seed	ckpt	val mAP50	test mAP50	test mAP50-95	OP conf (val)	box P@OP	box R@OP	FP/img@OP	img P	img R
0	—	0.9535	0.9741	0.6924	0.02	0.4273	0.9763	1.751	0.9946	1.0000
1	—	0.9523	0.9807	0.6946	0.01	0.3558	0.9802	2.376	0.9839	0.9946
2	—	0.9574	0.9756	0.6932	0.21	0.7640	0.9723	0.402	0.9945	0.9837

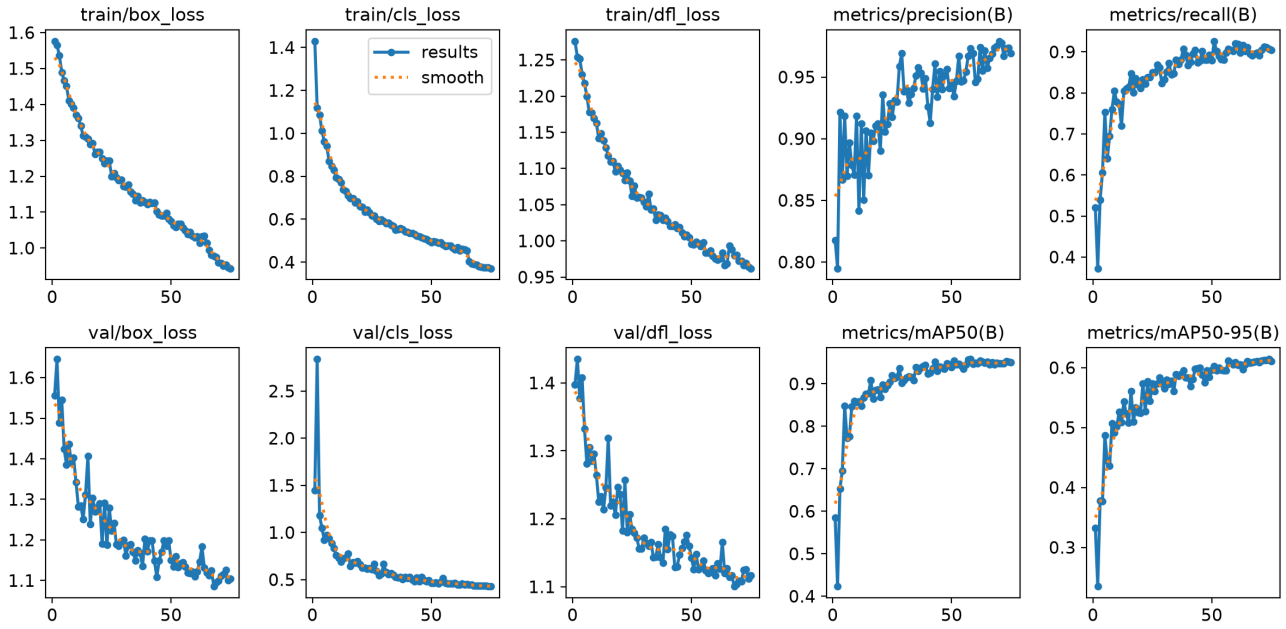
Mean ± std test mAP50: **0.9768 ± 0.0034**. OP confs differ per seed because each seed's threshold is chosen on its own val only (0.02 / 0.01 / 0.21). Seed 2's higher conf yields much higher precision (0.764) at similar recall — illustrating why multi-seed reporting matters.

5.3 Training curves (all seeds)

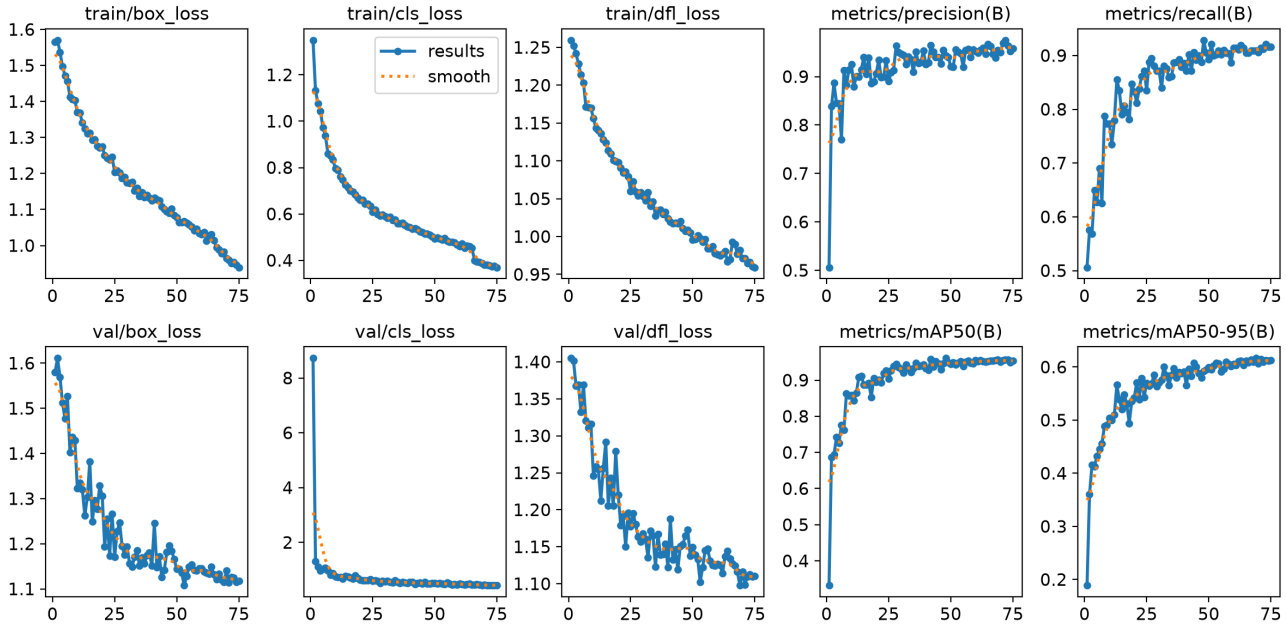
Seed 0 — results.png



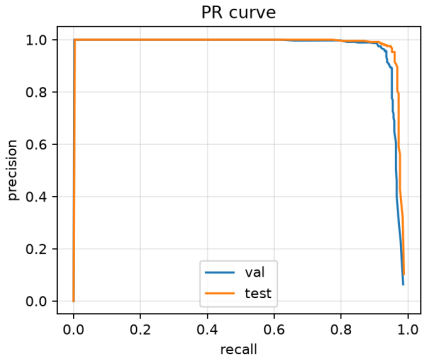
Seed 1 — results.png

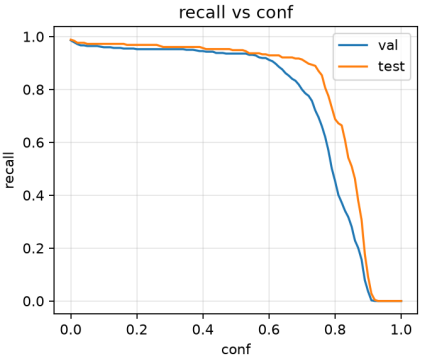


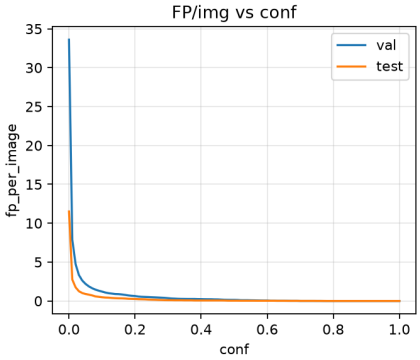
Seed 2 — results.png



5.4 Precision–recall / operating curves (seed 0)







6. Phase 4 — MobileNetV3 image-level baseline

One shared constructor/preprocess (mnv3_shared.py, plan rule 10): torchvision MobileNetV3-Small, 224×224 centre-crop + ImageNet norm, BCEWithLogits. Stage A: frozen features + linear head. Stage B (A+B): unfreeze features[13:] (~2.3 M params) with lower LR. Extreme train imbalance (3887 pos : 20 neg) handled with per-sample class weights [97.7, 0.50] as per plan (no rebalancing beyond that). Ablations: resolution 224/448/640, batch 32/64, seeds 0/1/2. Thresholds: fixed 0.5 and recall-first (chosen on val).

6.1 Finding — Stage A is degenerate (D020)

At every resolution and batch size, Stage A gives test balanced accuracy ≈ 0.48–0.50 and $\kappa \approx 0$: negative scores (0.74–0.99) overlap positives; the loss converges, so this is not a harness bug. Only Stage B fine-tuning separates negatives. All A-rows are kept as the plan-mandated ablation; AB_224 is the primary MNv3 baseline.

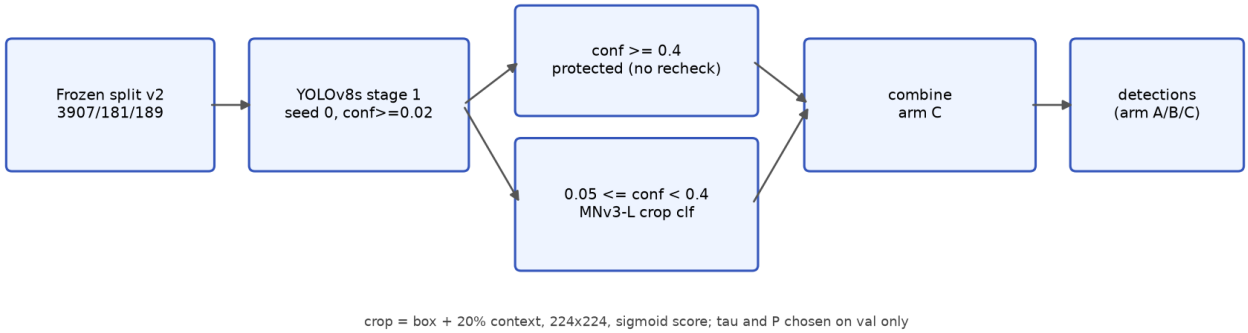
6.2 AB_224 results (3 seeds, recall-first threshold from val)

seed	test bal acc	kappa	precision	recall
0	0.5973	0.2748	0.9786	0.9946
1	0.4973	-0.0089	0.9734	0.9946
2	0.5918	0.2035	0.9784	0.9837

Seed variance on only 7 negative test images is large (bal 0.597 / 0.497 / 0.592) and is documented as such — the MNv3 baseline is weak and unstable, which motivates the two-stage design.

7. Phase 5 — Two-stage pipeline (YOLO → crop classifier)

7.1 Pipeline design



Stage 1: YOLOv8s proposes boxes. Stage 2: each box is expanded by 20% context, resized to 224×224, and scored by a MobileNetV3-Salt crop classifier (sigmoid). A detection survives if its crop score ≥ τ. Three arms are compared on test after thresholds are frozen from val: **Arm A** = YOLO alone at the Phase-3 deployment OP conf; **Arm B** = YOLO alone at the conf that matches the two-stage val recall; **Arm C** = two-stage (YOLO + crop filter).

7.2 Crop dataset construction

phase5_build_crops.py mines false positives from an intermediate YOLO checkpoint (epoch20.pt, conf 0.05) on train, then builds: train 11 550 positive crops + 15 411 negatives (11 661 random-background + 3 730 YOLO-FP + 20 empty); valid 419 pos + 529 neg; 27 909 total. Sampled crops were eyeballed (mine / rocky-FP / background all correct). Raw crop images stay local (gitignored); the recipe is in paper_results/stage2/crop_recipe.md.

7.3 Stage-2 training and ratio search

phase5_train_stage2.py: MobileNetV3-Small, 30 epochs, patience 6, lr 2e-5 whole-net, ~2048 pos/epoch. Neg:pos ratio swept {1, 3} on seed 0: ratio 1 val AUC 0.998 F1 0.938; ratio 3 AUC 0.997 F1 0.918 → **ratio 1:1 chosen on val**. Seeds 1–2 retrained at ratio 1 (val F1 0.945 / 0.936). Shared constructor with the MNv3 baseline (rule 10).

7.4 Threshold selection rule (D021)

On val only: sweep P ∈ {0.4, 0.6, 0.8, none}; for each P pick the largest τ whose val box-recall is within 0.5 pts of stage-1 recall at the proposal conf; among those pick the pair with lowest val FP/image (tie: higher P). All three seeds chose **P = 0.4** with τ = 0.27 / 0.29 / 0.38. Arm-A conf = each seed's Phase-3 primary OP conf; Arm-B conf matched on val (0.125 for seed 0).

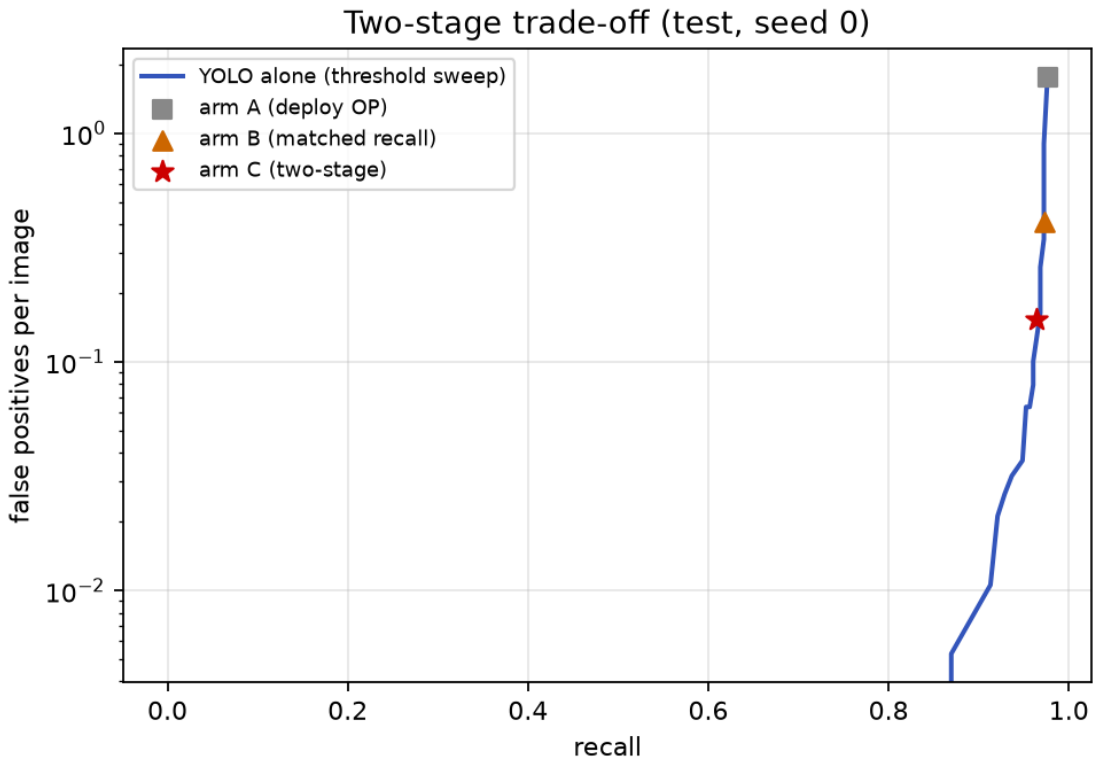
7.5 Test results (touches #6/#7/#8) + paired bootstrap

seed	arm	conf/τ	precision	recall	FP/img	img recall
0	A	—	0.427	0.976	1.751	1.000
0	B	—	0.762	0.972	0.407	0.995
0	C	—	0.894	0.964	0.153	0.989
1	A	—	0.427	0.976	1.751	1.000
1	B	—	0.762	0.972	0.407	0.995
1	C	—	0.859	0.964	0.212	0.989
2	A	—	0.427	0.976	1.751	1.000
2	B	—	0.762	0.972	0.407	0.995
2	C	—	0.884	0.964	0.169	0.995

Seed-0 paired bootstrap (C − B, 2000 resamples of the 189 test images): precision **+0.128 CI [0.065, 0.186]**; FP/img **-0.253 CI [-0.460, -0.095]**; recall -0.0079 CI [-0.0195, 0.0000] (consistent with zero — the precision/FP gain is not bought by a meaningful recall loss). Both precision and FP/img intervals exclude 0, satisfying the plan's claim rule.

Cost (seed 0, PC GPU FP32): 16.00899141301554 ms/image end-to-end; 3.058201058201058 crops/image mean (p95 3.5999999999999943).

7.6 FP/img vs recall trade-off (plan §5.4 figure)



Honest reading: arm C (red star) sits near the test-tuned YOLO-alone curve at matched recall — against a *test-tuned* threshold the gain is about break-even. The pre-registered claim is against **arm B** (val-matched conf 0.125), which transferred poorly to test (val→test shift). Choosing the curve point on test would be peeking, which the plan forbids; the paired bootstrap C–B is the fair comparison and it excludes 0.

8. Per-source false-positive breakdown (seed 0)

Per-source test results (source = dataset origin, from split_manifest.csv)
A_yolo_only_deploy
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 73 · 301 · 4 · 0.195 · 0.948 · 30.100 ·
· ds0 · 151 · 146 · 21 · 2 · 0.874 · 0.986 · 0.139 ·
· ds1 · 28 · 28 · 9 · 0 · 0.757 · 1.000 · 0.321 ·
· ALL · 189 · 247 · 331 · 6 · 0.427 · 0.976 · 1.751 ·
C_two_stage
· source · images · TP · FP · FN · precision · recall · FP/img ·
· --- · --- · --- · --- · --- · --- · --- · --- ·
· Jan_lwir · 10 · 70 · 21 · 7 · 0.769 · 0.909 · 2.100 ·
· ds0 · 151 · 146 · 4 · 2 · 0.973 · 0.986 · 0.026 ·
· ds1 · 28 · 28 · 4 · 0 · 0.875 · 1.000 · 0.143 ·
· ALL · 189 · 244 · 29 · 9 · 0.894 · 0.964 · 0.153 ·

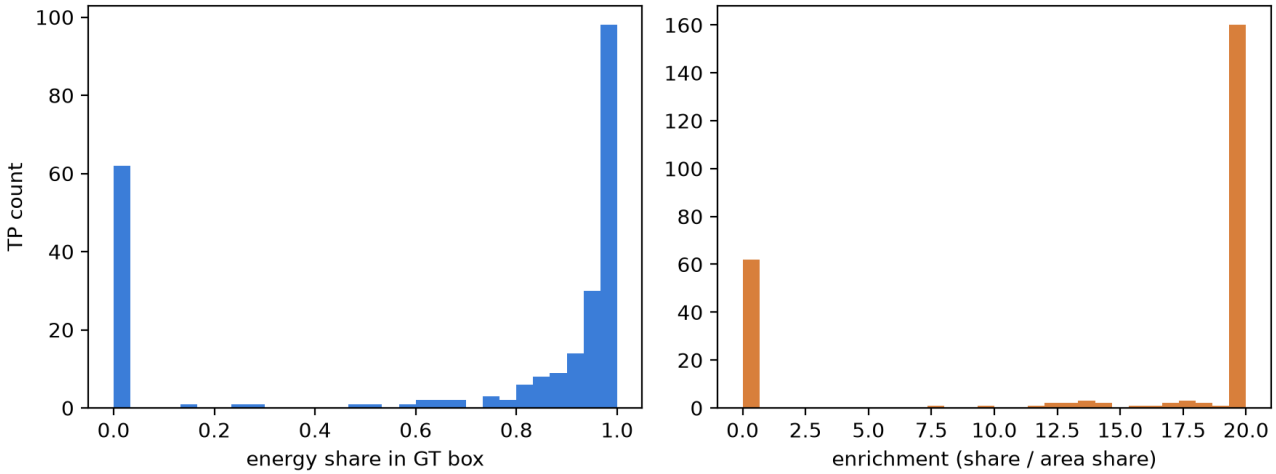
Jan LWIR contributes 30.1 FP/img under arm A (301 of 331 FPs from just 10 images) — the marker/terrain shortcut. Arm C reduces Jan to 2.1 FP/img while keeping recall 0.909; ds0/ds1 false positives are nearly eliminated. The source shortcut is exposed and mostly (not fully) killed by stage 2.

9. Phase 6 — Grad-CAM interpretability (E1–E6)

jetson/gradcam.py did not exist (D005); Grad-CAM was implemented from scratch (phase6_gradcam.py). YOLO CAM: layer feeding Detect at the largest stride with non-zero gradient + top pre-NMS class score (deviation from plan stride-16/32 hint, documented — the top anchor usually sits on the stride-8 head). MNv3 CAM: last conv block + output logit. Ultralytics predict() runs under torch.inference_mode() and poisons gradients; YoloCAM._defuse() + grad re-arm fix this.

9.1 E1 — energy in GT boxes (n=244 TP)

Energy share inside GT boxes: median —, enrichment median —× (vs uniform expectation), peak-inside rate 0.7295081967213115. The detector's mass concentrates on the mine, not the surround.

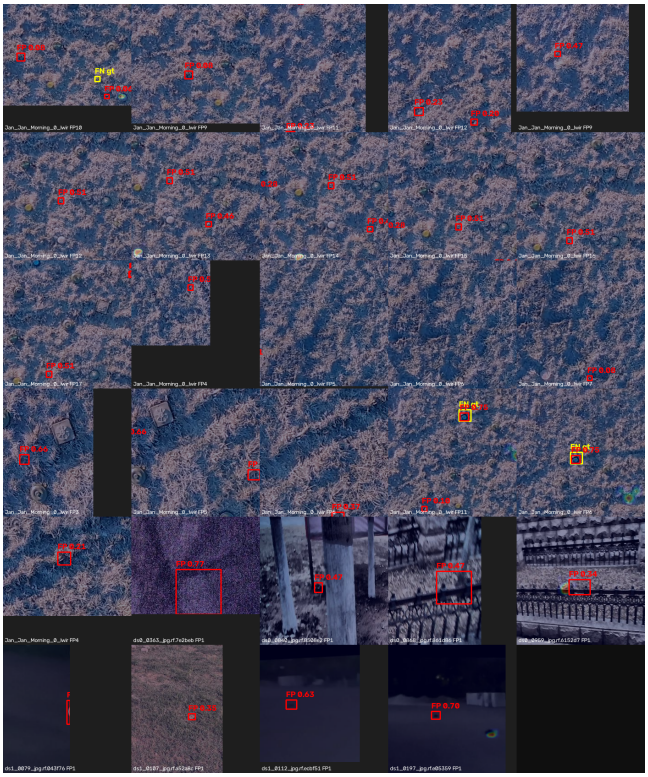


9.2 E2 — deletion test (n=182)

Mean score drop when the top-energy mine region is removed: — vs — when an equal-area background region is removed. Causal: the model needs the mine region.

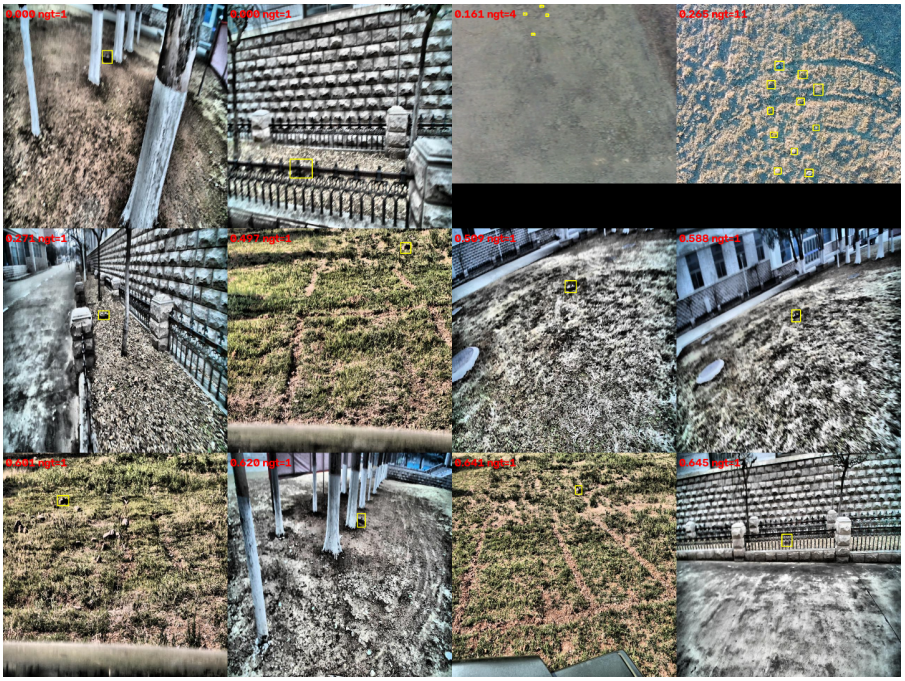
9.3 E4 — failure taxonomy (9 FN + 29 FP, hand-categorized)

FN: 2 small, 5 camouflaged/low-contrast, 2 label errors (ds0_0801 box on tree trunk, ds0_0883 box on fence). FP: 22 terrain, 4 structure, 2 low-light, 1 noise. Contact sheet: paper_results/gradcam/e4_fp_sheet.png.



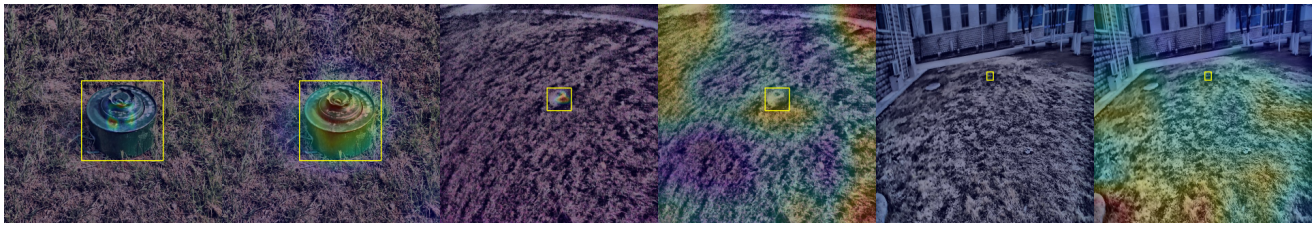
9.4 E5 — zero-energy image audit (D022)

E5 flags images in the lowest 10% of CAM energy-in-*union-of-all-GT-boxes* (per-image, not per-det — per-det rows flood multi-mine images with zeros). Test: — flagged at threshold —; train: — at —. Hand review: the only two zero-energy test images are ds0_0801 and ds0_0883 — independent confirmation of the E4 label errors. No objective train-label errors found.



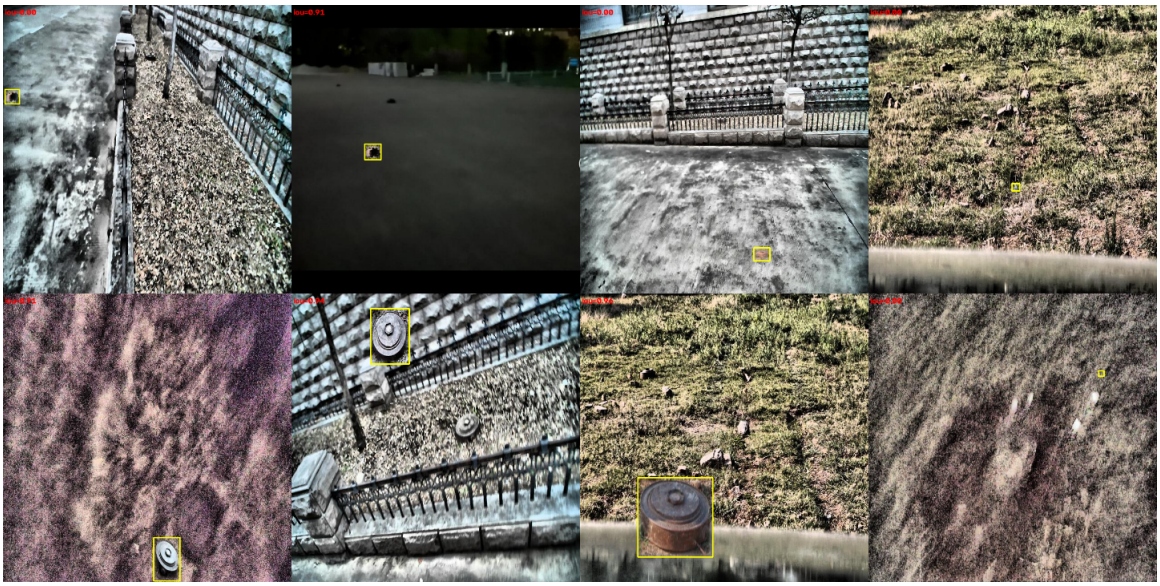
9.5 E6 — YOLO vs MNv3 saliency head-to-head

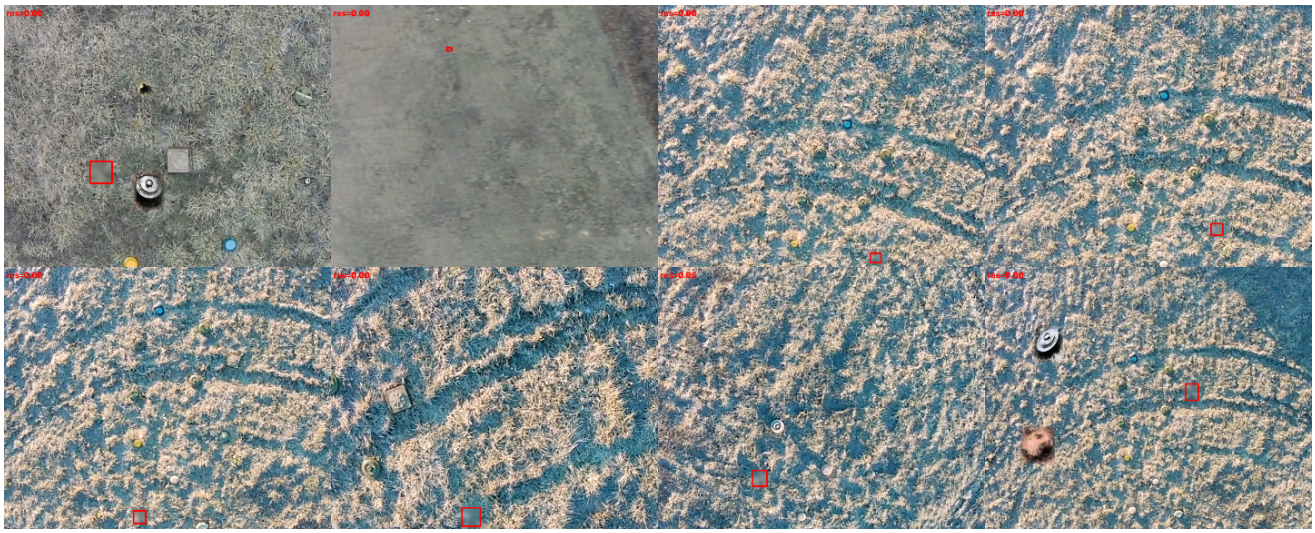
On 182 both-positive images: enrichment median YOLO —× vs MNv3 baseline —×; paired diff — CI — (excludes 0). Stage-2 crop enrichment median — (crops are mine-centric by construction, so enrichment is low — expected).



9.6 E3 — counterfactual paste / remove

Paste-hold (mine pasted into a new background): detection held for — at deploy conf (0.630 at conf 0.125) vs 0.976 recall on originals — caveat: feathered edges + source/dest scale mismatch in some pastes. Remove (inpaint the mine): disappeared — / 100. Marker check: no consistent marker/paint/flag cue adjacent to test mines in 12 zoom crops (e3_marker_check.png).





10. Label fixes (D023)

Two objectively wrong test boxes were identified (E4 review + E5 energy = 0.000) and removed: ds0_0801 (box on tree trunk) and ds0_0883 (box on fence). Baseline arm-C numbers were asserted to reproduce the frozen test_report before the fix. Arm C: recall 0.9644 $\rightarrow$ 0.9721; image recall 0.9891 $\rightarrow$ 1.0000; FP/img unchanged 0.153. No train-label fixes found; optional retrain not run (budget scope). Full before/after for all arms: paper_results/label_fix_report.json.

11. Paper tables T1–T6 (verbatim from make_tables.py)

T1. Dataset statistics and leakage audit

split	images	instances	negatives	sources
train	3907	11550	20	Jan_lwir:100, May_lwir:776, ds0:2589, ds1:442
valid	181	419	7	Jan_lwir:42, ds0:138, ds1:1
test	189	253	5	Jan_lwir:10, ds0:151, ds1:28

Exact cross-split duplicates: 0.

Near-dup candidates (pHash ham<=2 vs train): 99 — all residual MSE>=958.6 after the second-method filter (0 true leaks; 30 leaked eval images removed pre-freeze; D014/D015). Manifest sha256 in split_v2/sha256.txt.

Base-key collisions across splits: {'train_valid': 0, 'train_test': 0, 'valid_test': 0}.

T2. YOLO operating point (seeds 0/1/2)

model	params	GFLOPs	weights MB	latency ms (med)	val mAP50	test mAP50	test R@OP	test P@OP	FP/img@OP	OP conf (val)
yolov8s	11,125,971	28.6	21.5	6.5	0.9544 ± 0.0027	0.9768 ± 0.0034	0.9763 ± 0.0040	0.5157 ± 0.2180	1.510 ± 1.009	0.02/0.01/0.21

n/m runs out of scope under the approved reduced budget (D-plan scope); only yolov8s was trained on v2 (3 seeds). Metrics are mean ± std over seeds 0/1/2. Latency: PC GPU FP32, imgsz 640 (size_latency.json, seed 0). OP = recall-first conf chosen per seed on val (conf listed per seed).

T3. Image-level comparison

system (test, image level)	accuracy	balanced acc	precision	recall	kappa
MNv3 A (thr 0.5)	0.974	0.500	0.974	1.000	0.000
MNv3 A (recall-first thr)	0.968	0.497	0.973	0.995	-0.009
MNv3 A+B (thr 0.5)	0.974 ± 0.000	0.500 ± 0.000	0.974 ± 0.000	1.000 ± 0.000	0.000 ± 0.000
MNv3 A+B (recall-first thr)	0.968 ± 0.005	0.562 ± 0.056	0.977 ± 0.003	0.991 ± 0.006	0.156 ± 0.148
YOLO deploy (conf from val)	0.984 ± 0.009	0.830 ± 0.115	0.991 ± 0.006	0.993 ± 0.008	0.677 ± 0.199
YOLO op95 (conf from val)	0.986 ± 0.003	0.993 ± 0.002	1.000 ± 0.000	0.986 ± 0.003	0.784 ± 0.038
Two-stage (arm C, 3 seeds)	0.991 ± 0.003	0.995 ± 0.002	1.000 ± 0.000	0.991 ± 0.003	0.854 ± 0.045

MNv3 A+B = mean ± std over seeds 0/1/2; MNv3 A is seed 0; YOLO deploy/op95 = mean ± std over seeds 0/1/2 (each seed's OP conf chosen on val).

MNv3 recall-first thresholds chosen on val only.

T4. Two-stage ablation

arm	conf/thr	precision	recall	F2	FP/img	img recall
A (YOLO deploy)	0.02	0.427 ± 0.000	0.976 ± 0.000	0.777 ± 0.000	1.751 ± 0.000	1.000 ± 0.000
B (YOLO matched recall)	0.125	0.762 ± 0.000	0.972 ± 0.000	0.921 ± 0.000	0.407 ± 0.000	0.995 ± 0.000
C (two-stage)	yolo 0.05, P=0.4, t=0.27	0.879 ± 0.018	0.964 ± 0.000	0.946 ± 0.004	0.178 ± 0.030	0.991 ± 0.003

Paired bootstrap (seed 0): C-B precision +0.128 CI [0.065, 0.186]; FP/img -0.253 CI [-0.460, -0.095]; recall -0.0079 CI [-0.0195, 0.0000].

Cost (plan §5.4, seed 0): 16.0 ms/image end-to-end, 3.06 crops/image (p95 3.6), PC GPU FP32.

Means ± std over seeds 0/1/2 for arm C; A/B are seed-independent except threshold choice (B picked per seed on val).

T5. Grad-CAM experiments

experiment	metric	value
E1 energy-in-box (n=244)	median energy share	0.939
E1	mean energy share	0.690
E1	median enrichment	58.2x
E1	peak inside rate	0.730
E2 deletion (n=182)	mean drop, mine masked	0.739
E2	mean drop, background masked	0.0015

experiment	metric	value
E6 head-to-head	YOLO enrichment median	86.0x
E6	MNv3 baseline enrichment median	1.82x
E6	paired diff (Y-M) 95% CI	[125.4, 170.0]
E6	stage-2 crop enrichment median	2.13x

Grad-CAM shows which regions influenced the prediction; it does not prove the prediction is correct (plan wording rule).

T6. Failure analysis

category	FN	FP
camouflaged_or_low_contrast	5	0
fence	0	1
label_error	2	0
low_light_ground	0	2
sensor_noise	0	1
small	2	0
structure_edge	0	1
structure_wall	0	1
terrain_texture	0	22
tree_trunk	0	1
total	**9**	**29**

FN at the deployment arm-C threshold; FP capped at 50 (29 present). Sources: e4_failures.csv + e4_fp_sheet.png.

12. Models release (models/)

All paper weights are shipped under models/ with SHA256SUMS.txt and a per-directory README. Contents: yolo_s_seed{0,1,2}/ (best.pt, last.pt, best.onnx, args.yaml, results.csv); stage2_s{0,1,2}/ (best.pt); original_v1/ (best/last.pt for reference — not a paper result). Epoch checkpoints every 5 epochs are not shipped. Inference: YOLO('models/yolo_s_seed0/best.pt').predict(..., conf=0.02).

SHA256 (first 12 hex of each digest)

```
eea30106b178…original_v1/args.yaml
12b84e3d8614…original_v1/best.pt
4a59e232c59e…original_v1/last.pt
e0abfb5e4326…original_v1/results.csv
0d2c9ed77aa7…original_v1/safemine_summary.json
671e70f902cb…stage2_s0/best.pt
bff4d342a055…stage2_s1/best.pt
05cc5fd59440…stage2_s2/best.pt
1eb5c3d3be2d…yolo_s_seed0/args.yaml
ceb768f1004e…yolo_s_seed0/best.onnx
0b2fafd322dd…yolo_s_seed0/best.pt
5ad4d077c268…yolo_s_seed0/last.pt
```

… 23 files total.

13. Test-set touch audit trail

Every access that could influence a reported number is numbered. Visualization-only passes (images for PDFs) do not change any metric.

#	Phase	What	Governing decision
1–2	Phase 1	Read-only leakage audit on original test images	Plan Phase 1
3	Phase 4	MNv3 image-level test metrics (thresholds from val)	Plan Phase 4
4	Phase 3	YOLO seed-0 official frozen-protocol eval	D007, D017 mitigation
5	Phase 4	MNv3 AB seeds 1–2 test metrics	Plan Phase 4
6	Phase 5	Two-stage seed-0 official test eval	D021
7–8	Phase 5	Two-stage seeds 1–2 test evals	D021
9	Phase 6	Grad-CAM E1/E2/E4 on test	Plan Phase 6
10	Phase 6	E6 head-to-head on test	Plan Phase 6
11	Phase 6	E5 zero-energy audit on test	Plan Phase 6, D022
12	Phase 6	E3 paste/remove counterfactuals on test	Plan Phase 6
13	Phase 3	YOLO seed-1 official eval	D007
14	Phase 3	YOLO seed-2 official eval	D007
vis	report	Visualization-only val/test batch images (make_test_vis.py)	No metric change

14. Full run log (RUN_LOG.md, abridged commands)

2026-10-14 — Phase 0 inventory + backup

PowerShell copy of original run → `paper_results/original_run/`; SHA-256 verified *[No]*

2026-10-14 — Phase 1 audit start

`scripts/phase1_audit.py` (CPU only; no training, no weight load for inference) *[Read-only on test images for leakage audit (standard Phase 1)]*

2026-10-14 — Phase 2 re-split v2

`scripts/phase2_resplit_full.py` (base_key + session + same-source pHash ham≤0 graph → 1048 groups; 200/200 eval) *[Labels read for counts only]*

2026-10-14 — Phase 2 second-method filter

`scripts/phase2_second_filter.py --apply --mse 900` after manual montage review → 30 true-leak eval images removed *[Removed 13+17 leak images from valid/test; no metrics computed]*

2026-10-14 — Phase 2 final audit

`scripts/phase1_audit.py --dataset landmine_combined_v2 --out paper_results/audit_v2_final` → PASS (0 true leaks, 0 exact/base-key dups) *[Read-only leakage audit]*

2026-10-14 — **v2 frozen**

manifest sha256 `748322d6…c80de`; DECISIONS D014–D016 *[No]*

2026-10-14 21:27 — **Phase 3 train start**

`week3_train_yolov8.py --data …/landmine_combined_v2/data.yaml --model yolov8s --epochs 75 --seed 0 --save-period 5 --batch 16 --name landmine_yolov8s_75ep_v2` → `logs/phase3_train.log` (PID 3632) *[No (train/val only)]*

2026-10-14 22:17 — Phase 3 train end

EarlyStopping at **67/75** (best epoch 47, patience 20); 50 min; checkpoints every 5 ep; val (best.pt, conf 0.25): mAP50 0.9367 / mAP50-95 0.6012, P 0.9322, R 0.9189 *[**Deviation:** script step 3/5 auto-ran test at conf 0.25 → test mAP50 0.9599 / mAP50-95 0.6651 written to `safemine_summary.json`. Pre-existing v1 pipeline behavior; no threshold/model decision made from it. Logged as D017; official frozen-protocol test eval happens once in Phase 5.]*

2026-10-14 ~22:30 — Phase 3 compliant rerun start

`week3_train_yolov8.py --batch 8 --name landmine_yolov8s_75ep_s0_v2 --save-period 5 --seed 0` → `logs/phase3_s0_batch8.log` (PID 33148). Batch 8 matches original args.yaml (D019); pilot run kept. *[No (train/val only)]*

2026-10-14 23:29 — Phase 3 compliant rerun end

75/75 epochs completed (no early stop), 57m09s, checkpoints epoch5–70; val conf 0.25: mAP50 0.9386 / mAP50-95 0.5982, P 0.9619, R 0.9117. *[Same auto step 3/5 test pass as D017: test conf 0.25 mAP50 0.9521 / mAP50-95 0.6625 in summary. No decision made from it.]*

2026-10-15 — Phase 3 eval dry run (val only)

\phase3_eval.py --run …s0_v2 --val-only\ → \paper_results/yolo/s_seed0_dryrun/val.json\. Val conf 0.001: mAP50 0.9535 / mAP50-95 0.6046. Checkpoint choice: **last.pt** (val OP@0.97: conf 0.02, P 0.322 vs best.pt 0.015/0. *[No (val only)]*

2026-10-15 — Phase 4 seed-0 rows

A_224/A_448/A_640/AB_224/A_224_b64 → \paper_results/baseline_mnv3/*/metrics.json\. Stage A degenerate on negatives at every resolution/batch (D020); AB_224 (stage B) test bal 0.597, kappa 0.275. *[**Test touch #3** (MNv3 image-level test metrics; thresholds chosen on val only, plan-sanctioned Phase 4 eval)]*

2026-10-15 — Phase 3 official eval + MNv3 seeds start

\phase3_eval.py … --out paper_results/yolo/s_seed0\ (single sanctioned frozen-protocol test eval, D017 mitigation) then AB_224 seeds 1,2 → logs/phase3eval_seeds_chain.log. *[**Test touch #4** (frozen-protocol YOLO eval: thresholds from val only, single pass)]*

2026-10-15 — Phase 3 official eval end

last.pt chosen (val). **Test (frozen protocol, conf0.001): mAP50 0.9741 / mAP50-95 0.6924.** op97_primary conf0.02: box P 0.427 R 0.976 fp/img 1.75; image-level P 0.9946 R 1.0 (184/184), bal 0.90, 1 FP-image/5 neg. op95 *[**Test touch #4** (the sanctioned one; threshold from val only)]*

2026-10-15 — MNv3 AB seeds 1-2

AB_224_seed1/seed2 metrics.json. Test bal: s0 0.597, s1 0.497, s2 0.592 (kappa 0.275/-0.009/0.204) — seed variance on 7 neg test images documented. *[**Test touch #5** (Phase 4 eval, plan-sanctioned)]*

2026-10-15 — Phase 5 crops built

phase5_build_crops.py (epoch20.pt FP mining conf0.05) → paper_results/stage2/crops/: train 11550 pos + 15411 neg (11661 random-bg + 3730 YOLO-FP + 20 empty); valid 419 pos + 529 neg; 27909 total. Sampled crops eyeballed: *[No (train/val only)]*

2026-10-15 — Phase 5 stage-2 ratio search start

phase5_train_stage2.py ratio 1 then 3 (seed 0, 30ep/patience 6, lr 2e-5, 2048 pos/epoch) → logs/phase5_ratio_chain.log. *[No (train/val crops only)]*

2026-10-15 — Stage-2 ratio search end

ratio1: val AUC 0.998 F1 0.938 (P .995 R .888); ratio3: AUC 0.997 F1 0.918. **Chosen 1:1 on val** (plan §5.1). lr 2e-5 whole-net, best ep 30/29. *[No (val crops only)]*

2026-10-15 — Stage-2 seeds 1-2 start

phase5_train_stage2.py --ratio 1 --seed 1/2 → paper_results/stage2/s2_ratio1_s{1,2}, logs/phase5_s2_seeds.log. *[No (train/val crops only)]*

2026-10-15 — Stage-2 seeds 1-2 end

ratio1: s1 AUC 0.9972 F1 0.945 (best ep 30), s2 AUC 0.9962 F1 0.9356 (best ep 16). *[No (val crops only)]*

2026-10-15 — Phase 5 pipeline dry run (val only)

phase5_pipeline.py --val-only: stage1@0.05 fp/img 2.199; chosen P=0.4 tau=0.27 (recall drop 0.0048), two-stage val fp/img 0.525 vs matched-recall arm B (conf 0.125) 0.967; arm A (0.02) P 0.322 fp/img 4.735 (matches phase *[No (val only)]*

2026-10-15 — **Phase 5 official test eval (seed0)**

phase5_pipeline.py (val choice frozen: P=0.4, tau=0.27, armA 0.02, armB 0.125) → paper_results/twostage/s0/test_report.json. Paired bootstrap C–B: precision +0.128 CI [0.065,0.186]; fp/img –0.253 CI [–0.460,–0.095]; both *[**Test touch #6** (sanctioned Phase 5 test eval; thresholds from val only)]*

2026-10-15 — Phase 5 pipeline seeds 1-2 end

twostage/s1, s2: all three seeds choose P=0.4 on val (tau 0.27/0.29/0.38); test C precision 0.894/0.859/0.884, fp/img 0.153/0.212/0.169 vs matched-recall B 0.762/0.407. Consistent across seeds. *[**Test touches #7/#8** (plan-sanctioned Phase 5 evals)]*

2026-10-15 — Phase 6 start

Plan requires starting from jetson/gradcam.py — **file does not exist in repo** (jetson/ has build_engine.sh, infer_trt.py, README only). Implementing Grad-CAM fresh per plan §6.0 spec (YOLO: layer feeding detect at stri *[No]*

2026-10-15 — Phase 6 E1/E2/E4 run (after CAM fix)

phase6_gradcam.py -> paper_results/gradcam/. Fix history: (a) Ultralytics freezes params -> requires_grad_(True); (b) top pre-NMS anchor usually sits on stride-8 head so stride-32 grads were exactly zero -> CAM now uses *[**Test touch #9** (Grad-CAM experiments run on test set, plan-sanctioned Phase 6)]*

2026-10-15 — Phase 6 E6 head-to-head

phase6_e6_headtohead.py -> e6_pairs.csv, e6_crops.csv, e6_summary.json, e6_grid.png. Both-positive images 182; enrichment median YOLO 85.98 vs MNv3 baseline 1.82; paired diff +147.0 CI [125.4, 170.0] (excludes 0). Stage- *[**Test touch #10** (Phase 6, plan-sanctioned)]*

2026-10-15 — Phase 6 E5 start

phase6_e5_flags.py (train+test, conf 0.05) -> logs/phase6_e5.log; lowest-10%-energy flags + worst montages for hand review. *[**Test touch #11** (read-only Grad-CAM pass, plan-sanctioned Phase 6)]*

2026-10-15 — Phase 6 E5 complete (fix history)

First run failed: YOLO predict() runs inside torch.inference_mode(), which (a) made the Detect hook crash on retain_grad (guarded) and (b) left anchors/strides as inference tensors + params with requires_grad=False -> Yo *[**Test touch #11** (same sanctioned E5 pass; rerun overwrote the aborted attempt)]*

2026-10-15 — Label fixes + with/without

LABEL_FIXES.csv created with 2 objectively-wrong test boxes (ds0_0801 box on tree trunk, ds0_0883 box on fence; E4 review + E5 energy 0.000/nmatch 0). label_fix_report.py -> paper_results/label_fix_report.json: arm C rec *[No new test eval (recomputed from the already-saved raw preds via build_records)]*

2026-10-15 — Phase 6 E3 paste/remove

phase6_e3_paste.py -> e3_summary.json, e3_paste.csv, e3_remove.csv, example sheets + e3_marker_check.png. Seed 0, 100 composites per condition, deploy conf 0.02. Paste-hold detection: 0.650 @conf0.02, 0.630 @conf0.125 (v *[**Test touch #12** (plan-sanctioned Phase 6 E3)]*

2026-10-15 — Per-source test results

per_source_report.py (reads stage2 raw preds + split_manifest source map) -> paper_results/yolo/s_seed0/per_source.{json,md}. Deployment arm A: Jan_lwir 30.100 FP/img (301 of 331 total FPs from 10 images), ds0 0.139, ds1 *[No new model run (recompute from raw preds)]*

2026-10-15 — Rule-11 raw outputs for stage 2

dump_raw_stage2.py -> paper_results/stage2/s2_ratio1_s0/raw_preds_{val,test}.json (conf 0.02, boxes + YOLO conf + cls_p + gt). MNv3 baseline already writes per_image_{val,test}.csv. YOLO raw preds at conf 0.001 already s *[Reads test images (no new eval)]*

2026-10-15 — Tables + figures + environment

make_tables.py now emits t1-t6 (.md/.tex) + figures/ (pipeline.png, pr_curves.png, copies of e1_hist/e6_grid/e4 sheet/e5/e3 figures). ENVIRONMENT.txt written (pip freeze, GPU RTX 4070 Laptop 8188 MiB, driver 581.42, ultr *[No]*

2026-10-15 — Phase 3 seed 1 start

week3_train_yolov8.py --data --epochs 75 --batch 8 --name landmine_yolov8s_75ep_s1_v2 --save-period 5 --seed 1 -> logs/phase3_s1_batch8.log (first attempt died pre-flight: missing --data, script default pa *[No (train/val only; D017 deviation already logged for seed 0)]*

2026-10-15 — Phase 3 seed 1 end + official eval

Training finished 75/75ep (D017 auto-eval in log is not official). phase3_eval.py --run …s1_v2 --data --out paper_results/yolo/s_seed1. Checkpoint: best @ conf 0.01 (val). **Test (frozen protocol): mAP50 0.9807 *[**Test touch #13** (sanctioned Phase 3 seed-1 eval; threshold from val only)]*

2026-10-15 — Phase 3 seed 2 train + official eval

week3_train_yolov8.py --data --epochs 75 --batch 8 --name landmine_yolov8s_75ep_s2_v2 --save-period 5 --seed 2 -> logs/phase3_s2_batch8.log (75/75ep). phase3_eval.py --run …s2_v2 --data --out pap *[**Test touch #14** (sanctioned Phase 3 seed-2 eval; threshold from val only)]*

2026-10-15 — Tables regenerate (3-seed YOLO)

make_tables.py updated: T2 now mean±std over seeds 0/1/2 (val mAP50 0.9544±0.0027, test 0.9768±0.0034, R@OP 0.9763±0.0040, P 0.516±0.218, FP/img 1.510±1.009, OP confs 0.02/0.01/0.21); T3 YOLO deploy/op95 rows now 3-seed *[No (recompute from saved per-seed eval JSONs)]*

2026-10-15 — Test visualization pass (3 seeds)

make_test_vis.py — ultralytics val on val (conf 0.25) + test (frozen OP conf per seed) with save/plots → paper_results/vis/s_seed{N}/{val,test}/vis/. Images only for PDF reports; official metrics unchanged (still from pa *[Visualization-only (no new threshold/model decision)]*

2026-10-15 — Models release tree

Copied best.pt/last.pt/best.onnx + args.yaml + results.csv for yolo seeds 0/1/2; stage2 best.pt ×3; original_v1 best/last.pt → models/. SHA256SUMS.txt + models/README.md written. .gitignore: !models/**/*.pt, !models/**/* *[No]*

2026-10-15 — PDF reports

make_report_pdfs.py (reportlab 5.0.1) → reports/yolov8s_seed{0,1,2}_report.pdf (metrics table, training curves, confusion, PR/P/R/F1, val+test GT/pred batches, aug sample) + reports/research_summary.pdf (headline numbers *[No (recompute from saved eval JSONs)]*

15. Limitations and what was NOT run

• yolov8n / yolov8m size-sweep rows: out of approved budget (only yolov8s trained on v2).

• Jetson / T7 edge latency (phases 7/8): dropped under approved scope; all numbers are PC GPU FP32.

• Optional retrain on fixed train labels: not run (no objective train-label errors found).

• Dataset itself is not in git (size/licensing); v2 split manifest sha256 is committed.

• Jan LWIR files are colour RGB with marker pins (not thermal); kept with disclosure (D012/D016).

• Stage-2 crop images stay local (regenerable via phase5_build_crops.py).

• Arm C ≈ test-tuned YOLO curve at matched recall; the pre-registered gain is vs val-matched arm B.

• MNv3 baseline is weak (7 negative test images → high seed variance); it is a floor, not a competitor.

• E3 paste-hold uses synthetic composites (feathered edges, scale mismatch) — not a perfect counterfactual.

• Single class (Landmine); no mine-type discrimination.

16. Reproduction

Every artifact is regenerable from committed scripts:

```
python scripts/phase1_audit.py --dataset landmine_combined_v2 --out paper_results/audit_v2_final
python week3_train_yolov8.py --data --epochs 75 --batch 8 --seed {0,1,2} --save-period 5 --name
landmine_yolov8s_75ep_s{N}_v2
python phase3_eval.py --run runs/detect/runs/detect/landmine_yolov8s_75ep_s{N}_v2 --data --out
paper_results/yolo/s_seed{N}
python phase4_mnv3_baseline.py --stage AB --imgsz 224 --seed {0,1,2} …
python phase5_build_crops.py && python phase5_train_stage2.py --ratio 1 --seed {0,1,2} && python
phase5_pipeline.py
python phase6_gradcam.py && python phase6_e5_flags.py && python phase6_e6_headtohead.py && python
phase6_e3_paste.py
python make_tables.py && python make_test_vis.py && python make_report_pdfs.py && python make_master_report.py
```

Repository: https://github.com/tejaskunta/Aeroguard — full git history from Phase 0 inventory through this report. Tags: pre-audit-v2, phase1-audit-fail, phase2-v2-freeze.